as **implicit differentiation**. This process consists of differentiating both sides of an equation with respect to x, using the rules of differentiation and then solving for $\frac{dy}{dx}$. Since we think of y as being determined by the given equation as a differentiable function, the chain rule, in the form of the power rule for functions, gives the result.

$$\frac{d}{dx}(y^n) = n\,y^{n-1}\frac{dy}{dx}, \text{ where } n \text{ is an integer.}$$

Example 10.17

Find $\frac{dy}{dx}$ if $x^2 + y^2 = 1$.

Solution

We differentiate both sides of the equation,

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(1)$$

$$2x + 2y\frac{dy}{dx} = 0$$

Solving for the derivative yields

$$\frac{dy}{dx} = -\frac{x}{y}$$

Example 10.18

Find the slopes of the tangent lines to the graph of $x^2 + y^2 = 4$ at the points corresponding to $x = 1$.

Solution

Substituting $x = 1$ into the given equation yields $y^2 = 3$ or $y = \pm\sqrt{3}$.

Hence, there are tangent lines at $\left(1, \sqrt{3}\right)$ and $\left(1, -\sqrt{3}\right)$. Although $\left(1, \sqrt{3}\right)$ and $\left(1, -\sqrt{3}\right)$ are points on the graphs of two different implicit functions, we got the correct slope of each point. We have

$$\frac{dy}{dx} \text{ at } \left(1, \sqrt{3}\right) = -\frac{1}{\sqrt{3}} \text{ and } \frac{dy}{dx} \text{ at } \left(1, -\sqrt{3}\right) = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

Example 10.19

Find $\frac{dy}{dx}$ if $x^4 + x^2y^3 - y^5 = 2x + 1$.

Solution

Differentiating implicitly, we have

$$\frac{d}{dx}(x^4) + \frac{d}{dx}(x^2y^3) - \frac{d}{dx}(y^5) = \frac{d}{dx}(2x + 1)$$

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